Activity-5

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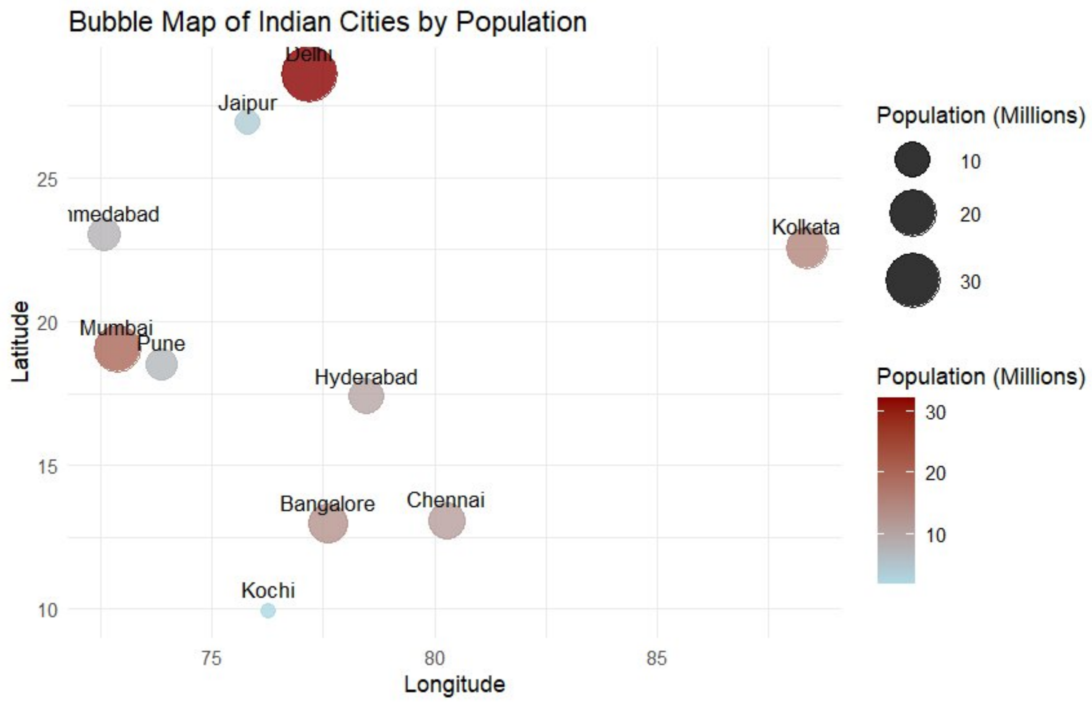
1. Geospatial Data Visualization

Dataset

City	Latitude	Longitude	Population (Millions)
Chennai	13.08	80.27	11
Bangalore	12.97	77.59	13
Hyderabad	17.38	78.48	10
Mumbai	19.07	72.87	20
Delhi	28.61	77.21	32
Pune	18.52	73.85	7
Kolkata	22.57	88.36	15
Ahmedabad	23.02	72.57	8
Jaipur	26.91	75.79	4
Kochi	9.93	76.26	2

Questions

```r
library(ggplot2)
city <- c("Chennai", "Bangalore", "Hyderabad", "Mumbai", "Delhi",
          "Pune", "Kolkata", "Ahmedabad", "Jaipur", "Kochi")
latitude <- c(13.08, 12.97, 17.38, 19.07, 28.61,
              18.52, 22.57, 23.02, 26.91, 9.93)
longitude <- c(80.27, 77.59, 78.48, 72.87, 77.21,
               73.85, 88.36, 72.57, 75.79, 76.26)
population <- c(11, 13, 10, 20, 32,
                7, 15, 8, 4, 2)
geo <- data.frame(city, latitude, longitude, population)
ggplot(geo, aes(x = longitude, y = latitude)) +
  geom_point(aes(size = population, color = population), alpha = 0.8) +
  geom_text(aes(label = city), vjust = -0.8, size = 3.5) +
  scale_size_continuous(range = c(3, 12)) +
  scale_color_gradient(low = "lightblue", high = "darkred") +
  labs(
    title = "Bubble Map of Indian Cities by Population",
    x = "Longitude",
    y = "Latitude",
    size = "Population (Millions)",
    color = "Population (Millions)"
  ) +
  theme_minimal()
```



5. Delhi has the highest population (32 million) and appears as the largest and darkest bubble on the map.

2. Network Visualization

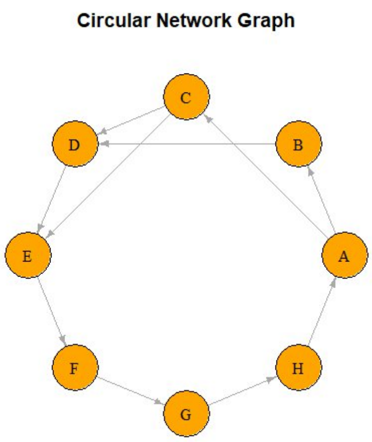
Dataset

From	To
A	B
A	C
B	D
C	D
C	E
D	E
E	F
F	G

From	To
G	H
H	A

Questions

```r
install.packages("igraph")
library(igraph)
from <- c("A", "A", "B", "C", "C", "D", "E", "F", "G", "H")
to   <- c("B", "C", "D", "D", "E", "E", "F", "G", "H", "A")
edges <- data.frame(from, to)
g <- graph_from_data_frame(edges, directed = TRUE)
plot(
  g,
  vertex.color = "skyblue",
  vertex.size = 30,
  vertex.label.color = "black",
  vertex.label.cex = 1,
  edge.arrow.size = 0.5,
  main = "Network Graph"
)
degree_values <- degree(g)
print(degree_values)
max_degree <- max(degree_values)
highest_node <- names(degree_values[degree_values == max_degree])
cat("Node(s) with highest degree:", highest_node,
    "\nDegree:", max_degree, "\n")
plot(
  g,
  layout = layout_in_circle(g),
  vertex.color = "orange",
  vertex.size = 30,
  vertex.label.color = "black",
  edge.arrow.size = 0.5,
  main = "Circular Network Graph"
)
```



3. A-3, B-2, C-3, D-3, E-3, F-2, G-2, H-2.

4. The nodes with the highest degree are A, C, D, and E, each having 3 connections.

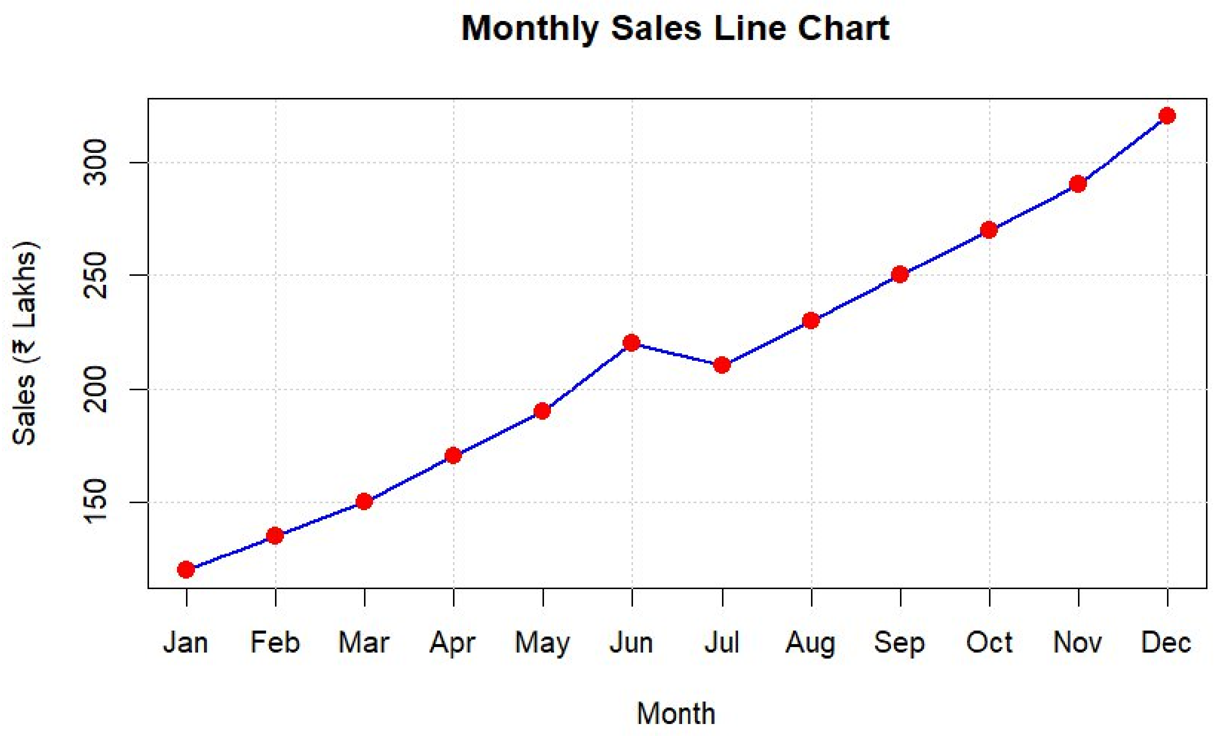
3. Temporal Data Representation

Dataset

Month	Sales (₹ Lakhs)
Jan	120
Feb	135
Mar	150
Apr	170
May	190
Jun	220
Jul	210
Aug	230
Sep	250
Oct	270
Nov	290
Dec	320

Questions

```
month <- c("Jan", "Feb", "Mar", "Apr", "May", "Jun",
           "Jul", "Aug", "Sep", "Oct", "Nov", "Dec")
sales <- c(120, 135, 150, 170, 190, 220,
           210, 230, 250, 270, 290, 320)
plot(
  sales,
  type = "o",
  pch = 19,
  xaxt = "n",
  col = "blue",
  lwd = 2,
  xlab = "Month",
  ylab = "Sales (₹ Lakhs)",
  main = "Monthly Sales Line Chart"
)
axis(1, at = 1:12, labels = month)
grid()
points(1:12, sales, pch = 19, col = "red", cex = 1.3)
```



5. December has the maximum sales, with ₹320 lakhs, and appears as the highest point on the line chart.

4. Multivariate Data Visualization

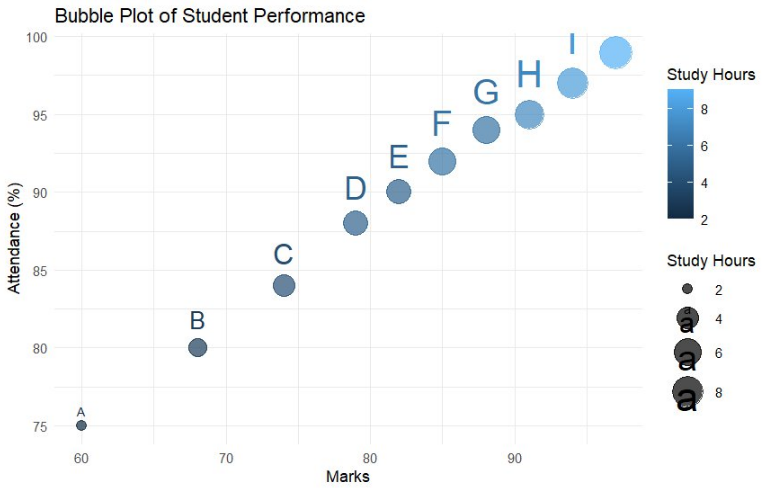
Dataset

Student	Marks	Attendance (%)	Study Hours	Projects
A	60	75	2	1
B	68	80	3	2
C	74	84	4	2
D	79	88	5	3
E	82	90	5	3
F	85	92	6	4
G	88	94	6	4
H	91	95	7	5
I	94	97	8	5
J	97	99	9	6

Questions

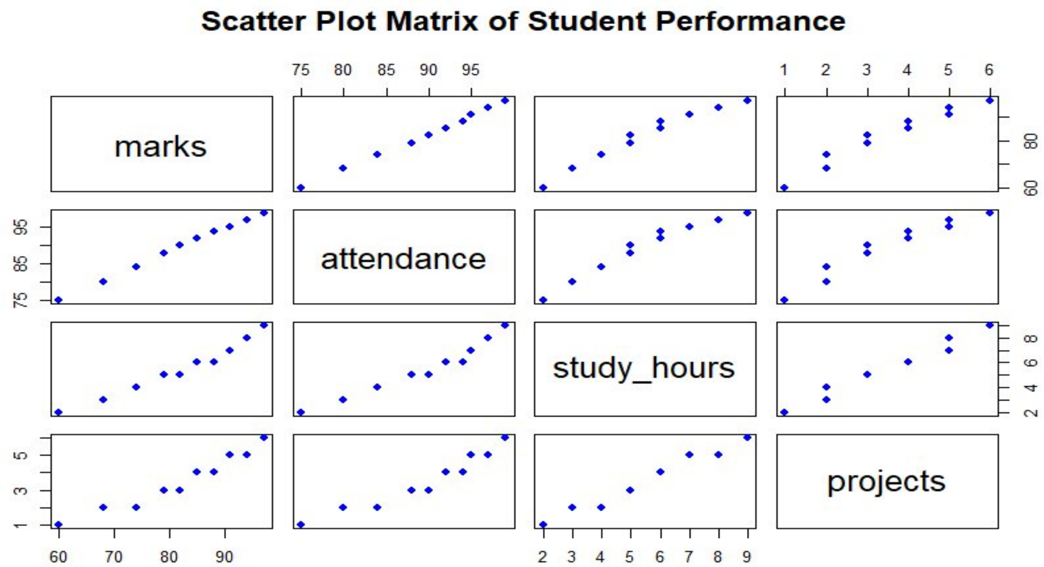
1

```r
library(ggplot2)
student <- c("A","B","C","D","E","F","G","H","I","J")
marks <- c(60,68,74,79,82,85,88,91,94,97)
attendance <- c(75,80,84,88,90,92,94,95,97,99)
study_hours <- c(2,3,4,5,5,6,6,7,8,9)
projects <- c(1,2,2,3,3,4,4,5,5,6)
data <- data.frame(student, marks, attendance, study_hours, projects)
ggplot(data, aes(x = marks, y = attendance,
                 size = study_hours, color = study_hours)) +
  geom_point(alpha = 0.7) +
  geom_text(aes(label = student), vjust = -1) +
  scale_size(range = c(3, 10)) +
  labs(
    title = "Bubble Plot of Student Performance",
    x = "Marks",
    y = "Attendance (%)",
    size = "Study Hours",
    color = "Study Hours"
  ) +
  theme_minimal()
```



2.

```r
student <- c("A","B","C","D","E","F","G","H","I","J")
marks <- c(60,68,74,79,82,85,88,91,94,97)
attendance <- c(75,80,84,88,90,92,94,95,97,99)
study_hours <- c(2,3,4,5,5,6,6,7,8,9)
projects <- c(1,2,2,3,3,4,4,5,5,6)
data <- data.frame(student, marks, attendance, study_hours, projects)
pairs(
  data[, c("marks", "attendance", "study_hours", "projects")],
  main = "Scatter Plot Matrix of Student Performance",
  pch = 19,
  col = "blue"
)
```



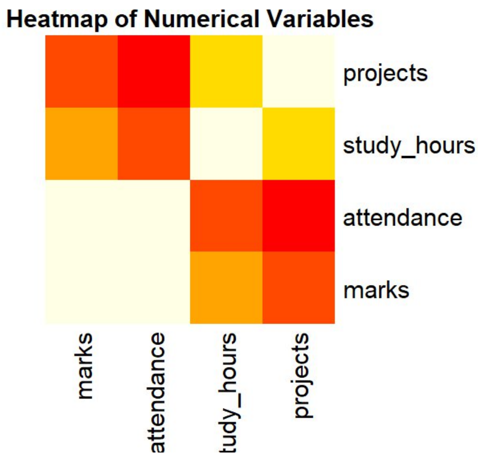
3.

```
              marks attendance study_hours
marks       1.0000000  0.9984669   0.9814683
attendance  0.9984669  1.0000000   0.9735953
study_hours 0.9814683  0.9735953   1.0000000
```

4. Student J has the highest marks (97) and also has the largest bubble, as the bubble size represents Study Hours (9 hours).

5.

```r
num_data <- data[, c("marks", "attendance", "study_hours", "proj
cor_matrix <- cor(num_data)
heatmap(
  cor_matrix,
  Rowv = NA,
  Colv = NA,
  col = heat.colors(20),
  scale = "none",
  margins = c(8, 8),
  main = "Heatmap of Numerical Variables"
)
```



5. Integrated Data Visualization Exercise

Dataset

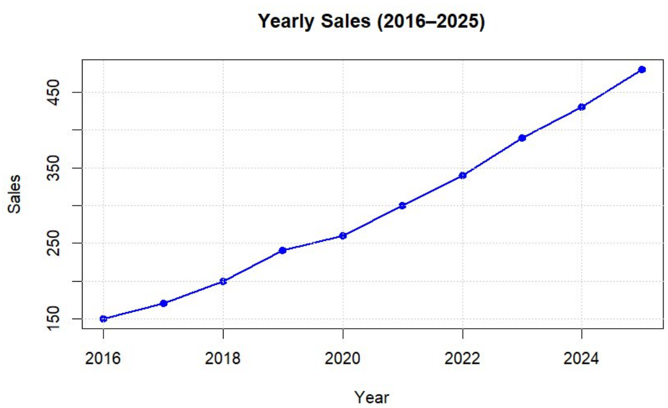
Year	Sales	Profit	Customers	Branches
2016	150	20	1200	5
2017	170	25	1500	6
2018	200	32	1800	7
2019	240	40	2200	8
2020	260	42	2400	8
2021	300	55	2800	10
2022	340	65	3300	11
2023	390	72	3800	12
2024	430	82	4200	13
2025	480	95	4700	15

Questions

1.

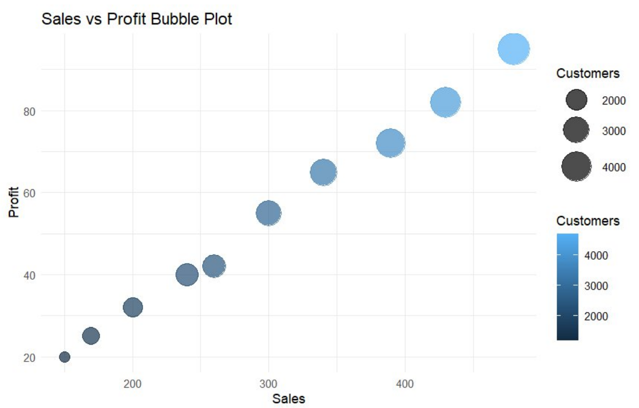
```r
year <- c(2016, 2017, 2018, 2019, 2020,
          2021, 2022, 2023, 2024, 2025)
sales <- c(150, 170, 200, 240, 260,
           300, 340, 390, 430, 480)
plot(
  year, sales,
  type = "o",
  pch = 19,
  col = "blue",
  lwd = 2,
  xlab = "Year",
  ylab = "Sales",
  main = "Yearly Sales (2016-2025)"
)

grid()
```



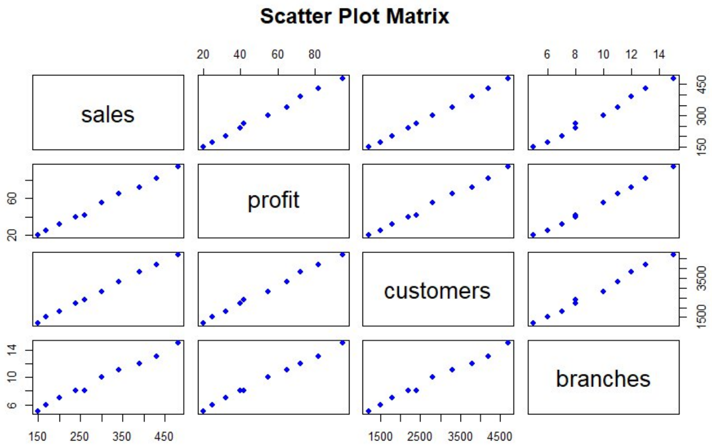
2.

```r
year <- c(2016, 2017, 2018, 2019, 2020,
          2021, 2022, 2023, 2024, 2025)
sales <- c(150, 170, 200, 240, 260,
           300, 340, 390, 430, 480)
profit <- c(20, 25, 32, 40, 42,
            55, 65, 72, 82, 95)
customers <- c(1200, 1500, 1800, 2200, 2400,
               2800, 3300, 3800, 4200, 4700)
branches <- c(5, 6, 7, 8, 8,
              10, 11, 12, 13, 15)
data <- data.frame(year, sales, profit, customers, branches)
library(ggplot2)
ggplot(data, aes(x = sales, y = profit,
                 size = customers, color = customers)) +
  geom_point(alpha = 0.7) +
  scale_size(range = c(4, 12)) +
  labs(
    title = "Sales vs Profit Bubble Plot",
    x = "Sales",
    y = "Profit",
    size = "Customers",
    color = "Customers"
  ) +
  theme_minimal()
```



4.

```r
year <- c(2016, 2017, 2018, 2019, 2020,
          2021, 2022, 2023, 2024, 2025)
sales <- c(150, 170, 200, 240, 260,
           300, 340, 390, 430, 480)
profit <- c(20, 25, 32, 40, 42,
            55, 65, 72, 82, 95)
customers <- c(1200, 1500, 1800, 2200, 2400,
               2800, 3300, 3800, 4200, 4700)
branches <- c(5, 6, 7, 8, 8,
              10, 11, 12, 13, 15)
data <- data.frame(year, sales, profit, customers, branches)
pairs(
  data[, c("sales", "profit", "customers", "branches")],
  main = "Scatter Plot Matrix",
  pch = 19,
  col = "blue"
)
```



5. As the number of branches increases, both sales and profits increase, indicating a strong positive relationship.